Mock Test — Solutions

C Programming Fundamentals: 25 questions, worked step by step

CPE101 Teaching Assistant

CPE101 Computer Programming
King Mongkut's University of Technology Thonburi (KMUTT)

Tutorial session

How to use these slides

- 25 questions $\times$ 4 marks, grouped by topic (not in exam order).
- Each question appears first. **Try it yourself**, then click to reveal the answer.
- Every program here was compiled with `gcc -Wall -Wextra -std=c11 … -lm` and every output shown is from a **real run**.
- Source code: `03-mock-solutions/examples/qNN_*.c`

TIP Try it yourself before revealing the answer, then compare with the reasoning on the slide.

Topic	Questions
Program structure	1, 2
Output and input	3, 5, 8, 9, 10
Types and constants	4, 6, 7
Formulas	11, 12
Decisions	13, 14, 15, 16
Loops	17 – 22
Operators, arrays	23, 24
Series	25

Q1 · The basic structure of a C program

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Definition	`#define PI 3.14159`	symbolic constants, replaced by the preprocessor
Global declaration	global variables, function prototypes	names usable by **every** function in the file
`main()`	`int main(void) { ... }`	**required**; execution starts and ends here
Sub-program	user-defined functions	smaller tasks called from `main()`

Q1 · One program with all six sections

```c
/*  Documentation section
 *  Program : Area of a circle
 *  Author  : CPE101 TA
 */
#include <stdio.h>          /* Link section       */

#define PI 3.14159          /* Definition section */

double area(double r);      /* Global declaration */
int call_count = 0;

int main(void)              /* main() section     */
{
    double r = 2.0;
    printf("Area = %.2f\n", area(r));
    printf("area() was called %d time(s)\n", call_count);
    return 0;
}

double area(double r)       /* Sub-program section */
{
    call_count++;
    return PI * r * r;
}
```

```
Area = 12.57
area() was called 1 time(s)
```

- Only `main()` is compulsory. The others are used when needed.
- The prototype in the global section lets `main()` call `area()` **before** its body appears.
- `call_count` is global, so both functions can use it.

TIP Every program must have `main()`. Include the other parts only as needed.

Q2 · Multi-file programs, `extern` and `static`

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- **`extern int total;`**: "`total` is *defined in another file*". No memory is created. Exactly **one** file has the real definition `int total = 0;`. Functions are `extern` by default, so a prototype is enough.

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- **Header file** (`.h`): holds *declarations*, i.e. prototypes and `extern` variables. Every `.c` file that `#includes` it agrees on the same names and types. Protect it with an include guard.
- **`extern int total;`**: "`total` is *defined in another file*". No memory is created. Exactly **one** file has the real definition `int total = 0;`. Functions are `extern` by default, so a prototype is enough.
- **`static`** on a global variable or function: **internal linkage**. The name is visible *only inside its own file*. Another file that tries to use it gets a link error (`undefined reference`).

Q2 · Example

counter.h

```h
#ifndef COUNTER_H
#define COUNTER_H

extern int total;
void add(int x);

#endif
```

counter.c

```c
#include "counter.h"

int total = 0;         /* definition */
static int calls = 0; /* private    */

void add(int x)
{
    calls++;
    total += x;
}
```

main.c

```c
#include <stdio.h>
#include "counter.h"

int main(void)
{
    add(5);
    add(7);
    printf("total = %d\n", total);
    /* calls is NOT usable here */
    return 0;
}
```

```
$ gcc -Wall -Wextra -std=c11 main.c counter.c -o q02
$ ./q02
total = 12
```

TIP `extern` = “exists in another file”; `static` = “usable only in this file” (see the Compiler slides for details).

Q3 · A pattern with `printf()` only

Question. Write a C program that uses only `printf()` statements to display the pattern (two centred lines): `WELCOME  TO / C  LAB`

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```c
#include <stdio.h>

int main(void)
{
    printf("WELCOME  TO\n");
    printf("  C  LAB\n");
    return 0;
}
```

```
WELCOME  TO
  C  LAB
```

- Each line ends with `\n`.
- `WELCOME  TO` is 11 characters and `C  LAB` is 6, so we need $(11-6)/2 \approx 3$ leading spaces to centre it.
- Keep the **double spaces** exactly as shown.

Q3 · A pattern with `printf()` only

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- Keep the **double spaces** exactly as shown.

Tip: can't see how many spaces you printed? Temporarily print `*` instead: `"***C**LAB"`.

Q5 · Escape sequences: what and why

Question. What is an escape sequence? Why does C need escape sequence characters? Explain any five escape sequences (e.g. `\n`, `\t`, `\b`, `\"`, `\a`) with one `printf()` example each.

TIP Escape sequences stand in for characters you cannot type directly in a string, such as a newline or the " character.

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- **Why needed?** Some characters can't be typed directly inside "...":
  - **invisible / control** characters: new line, tab, backspace, beep
  - characters that have a **special meaning** in a string: " ends the string, \ starts an escape

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- **Why needed?** Some characters can't be typed directly inside "...":
  - **invisible / control** characters: new line, tab, backspace, beep
  - characters that have a **special meaning** in a string: " ends the string, \ starts an escape
- The compiler turns `\n` into the single new-line character (ASCII 10). `"A\n"` is 2 characters, not 3.

TIP Escape sequences stand in for characters you cannot type directly in a string, such as a newline or the " character.

Q5 · Five (+1) escape sequences in action

```c
printf("Line 1\nLine 2\n");     /* \n  new line       */
printf("Name\tScore\n");        /* \t  horizontal tab */
printf("ABC\bD\n");             /* \b  backspace      */
printf("He said \"Hi\"\n");     /* \"  double quote   */
printf("Done!\a\n");            /* \a  alert (beep)   */
printf("C:\\cpe101\\lab1\n");   /* \\  backslash      */
```

\n	move to the start of the next line
\t	jump to the next tab stop
\b	move the cursor back one place
\”	a literal ” inside a string
\a	ring the terminal bell (nothing visible)
\\	a literal backslash

On screen:

```
Line 1
Line 2
Name    Score
ABD
He said "Hi"
Done!
C:\cpe101\lab1
```

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On screen:

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Line 1
Line 2
Name    Score
ABD
He said "Hi"
Done!
C:\cpe101\lab1
```

ABC\bD: the cursor steps back over C, then D is written on top of it ⇒ ABD.

\a makes a beep (if sound is on), but prints nothing.

Q8 · scanf with field widths

Question. `scanf("%2d%*1d%3d%d", &a, &b, &c);` The user enters `9876543210`. Find a, b, c. Explain the `*` in `%*1d`. What if the input were `98 7 654 3`?

9	8	7	6	5	4	3	2	1	0

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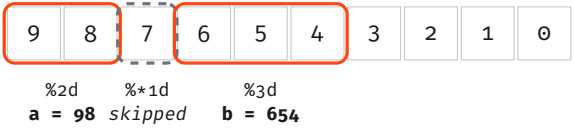
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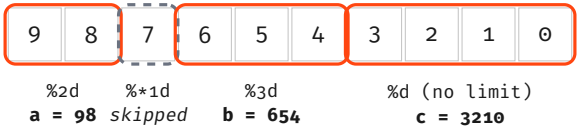
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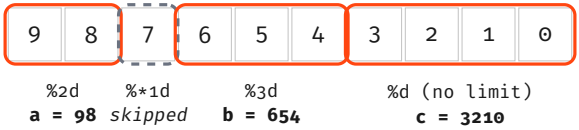
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- A number in `%2d` is the **maximum width**: read at most 2 characters.
- `*` = **assignment suppression**: read the field, then throw it away. It needs **no** variable in the list.

```
a = 98, b = 654, c = 3210
```

Q8 · And with spaces: 98 7 654 3

%d skips any **whitespace** (spaces, tabs, newlines) *before* it starts reading. The width then counts only digits.

Spec	Reads	Result
%2d	98	a = 98

Q8 · And with spaces: 98 7 654 3

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`%2d`	`98`	`a = 98`
`%*1d`	skips space, reads `7`	discarded

Q8 · And with spaces: 98 7 654 3

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Spec	Reads	Result
`%2d`	`98`	`a = 98`
`%*1d`	skips space, reads `7`	discarded
`%3d`	skips space, reads `654`	`b = 654`

Q8 · And with spaces: 98 7 654 3

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Spec	Reads	Result
%2d	98	a = 98
%*1d	skips space, reads 7	discarded
%3d	skips space, reads 654	b = 654
%d	skips space, reads 3	c = 3

```
$ echo "98 7 654 3" | ./q08_scanf
a = 98, b = 654, c = 3
```

Same a and b. Only c changes: `3210` → `3`, because %d stops at the end of the input.

Here the spaces happen to match the field widths. A width never makes `scanf` read *past* a space inside a number.

TIP `%2d` = read at most 2 digits; `%*1d` = read 1 digit and discard it; `%d` skips leading whitespace automatically.

Q9 · `getchar()` / `putchar()`: vowels and spaces

Question. Use `getchar()` to read characters until a full stop (`.`). Count (a) vowels and (b) spaces. Echo each character in uppercase with `putchar()`.

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1. Loop: read one character with `ch = getchar()`; stop when `ch == '.'` (also stop at EOF, end of input).

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2. If `ch` is a vowel (upper or lower case) $\rightarrow$ `vowels++`. If `ch == ' '` $\rightarrow$ `spaces++`.

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2. If `ch` is a vowel (upper or lower case) $\rightarrow$ `vowels++`. If `ch == ' '` $\rightarrow$ `spaces++`.
3. Echo it: `putchar(toupper(ch))`. `toupper` is in `<ctype.h>` and leaves non-letters unchanged.

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2. If `ch` is a vowel (upper or lower case) → `vowels++`. If `ch == ' '` → `spaces++`.
3. Echo it: `putchar(toupper(ch))`. `toupper` is in `<ctype.h>` and leaves non-letters unchanged.
4. After the loop, print both counts.

TIP `getchar()` reads one character at a time (spaces included), unlike `scanf("%s")`, which stops at whitespace.

Q9 · Solution

```c
#include <stdio.h>
#include <ctype.h>      /* tolower, toupper */
int main(void)
{
    int ch, vowels = 0, spaces = 0;
    printf("Type a sentence ending with '.':\n");
    while ((ch = getchar()) != '.' && ch != EOF)
    {
        switch (tolower(ch))
        {
            case 'a': case 'e': case 'i':
            case 'o': case 'u':
                vowels++;
                break;
            case ' ':
                spaces++;
                break;
        }
        putchar(toupper(ch));
    }
    printf("\nVowels: %d\nSpaces: %d\n", vowels, spaces);
    return 0;
}
```

```
Type a sentence ending with '.':
Hello World, I love C. extra
HELLO WORLD, I LOVE C
Vowels: 6
Spaces: 4
```

- `ch` is an `int`, because `getchar()` returns `int` (so it can also return `EOF`).
- `tolower` lets one `case` list catch `'A'` and `'a'`.
- Text after the `.` is never read.
- Check: e, o, o, l, o, e = 6 vowels.

Q10 · Format specifiers, exactly

Question. `int n = 42; float f = 3.14159;` Write the exact output (use _ for a space) and explain each format specifier.

Statement	Output	Why
`printf("[%5d]", n);`	`[___42]`	width 5, **right**-aligned: 3 spaces + 2 digits

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`printf("[%5d]", n);`	`[___42]`	width 5, **right**-aligned: 3 spaces + 2 digits
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`printf("[%.2f]", f);`	`[3.14]`	2 digits after the point (rounded)

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`printf("[%.2f]", f);`	`[3.14]`	2 digits after the point (rounded)
`printf("[%8.3f]", f);`	`[___3.142]`	total width 8 *including* the point: `3.142` is 5 chars, + 3 spaces. Rounded: `3.14159` → `3.142`

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`printf("[%-5d]", n);`	`[42___]`	- means **left**-aligned: padding goes after
`printf("[%.2f]", f);`	`[3.14]`	2 digits after the point (rounded)
`printf("[%8.3f]", f);`	`[___3.142]`	total width 8 *including* the point: `3.142` is 5 chars, + 3 spaces. Rounded: `3.14159` → `3.142`
`printf("[%e]", f);`	`[3.141590e+00]`	scientific notation, 6 digits after the point by default: 3.14159×10^{0}

TIP `%8.3f` = field width 8, counting the decimal point; `%-5d` = left-aligned.

Q4 · Basic data types

Question. What are the basic data types in C? Give the typical size and range of each. Describe how `short`, `long`, `signed`, `unsigned` change these ranges, with one example each.

Type	Typical size	Range	Used for
`char`	1 byte	-128 to 127	one character / small integer

Q4 · Basic data types

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Type	Typical size	Range	Used for
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`int`	4 bytes	$-2\,147\,483\,648$ to $2\,147\,483\,647$	whole numbers

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Type	Typical size	Range	Used for
`char`	1 byte	-128 to 127	one character / small integer
`int`	4 bytes	$-2\,147\,483\,648$ to $2\,147\,483\,647$	whole numbers
`float`	4 bytes	about $\pm 3.4 \times 10^{38}$, $\approx$ 6–7 significant digits	real numbers

Q4 · Basic data types

Question. What are the basic data types in C? Give the typical size and range of each. Describe how `short`, `long`, `signed`, `unsigned` change these ranges, with one example each.

Type	Typical size	Range	Used for
`char`	1 byte	-128 to 127	one character / small integer
`int`	4 bytes	$-2\,147\,483\,648$ to $2\,147\,483\,647$	whole numbers
`float`	4 bytes	about $\pm 3.4 \times 10^{38}$, $\approx$ 6–7 significant digits	real numbers
`double`	8 bytes	about $\pm 1.7 \times 10^{308}$, $\approx$ 15–16 significant digits	precise real numbers

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`double`	8 bytes	about $\pm 1.7 \times 10^{308}$, $\approx$ 15–16 significant digits	precise real numbers
`void`	—	no value	"returns nothing", `main(void)`

"Typical" matters: the C standard only guarantees *minimum* ranges (e.g. `int` at least 16 bits). Check with `sizeof` on your machine.

Q4 · Qualifiers: `short`, `long`, `signed`, `unsigned`

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Real sizes on Windows (MSYS2 gcc), from `q04_sizes.c`:

```
char      1 byte   -128 .. 127
short     2 bytes  -32768 .. 32767
int       4 bytes  -2147483648 .. 2147483647
unsigned  4 bytes  0 .. 4294967295
long      4 bytes  -2147483648 .. 2147483647
long long 8 bytes  -9223372036854775808 ..
                    9223372036854775807
float     4 bytes  up to 3.402823e+38 (6 digits)
double    8 bytes  up to 1.797693e+308 (15 digits)
```

Note: `long` is 4 bytes on Windows but **8 bytes** on 64-bit Linux/macOS.

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```
int count;
```

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- **Declare**: give a name and type; memory is reserved.
  ```
  int count;
  ```
- **Initialize**: give it a first value *at the moment it's declared*.
  ```
  int total = 0;
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  `int total = 0;`
- **Garbage value**: an uninitialized local variable holds whatever bits were already in that memory. Unpredictable, and different from run to run.

```c
int count;          /* garbage   */
int total = 0;      /* initialized */
for (int i = 1; i <= 3; i++)
{
    count = count + 1;
    total = total + i;
}
printf("count = %d (expected 3)\n", count);
printf("total = %d (expected 6)\n", total);
```

```
count = 32761 (expected 3)
total = 6 (expected 6)
```

`count` started from garbage, so the sum is meaningless.

TIP Always initialise accumulators (`sum = 0`, `product = 1`), or you will get garbage values.

Q7 · Symbolic constants

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- **Advantage 2 – easy to change**: if VAT becomes 10%, edit **one** line, not every place 0.07 appears (and you can't miss one).
- Also: the value can't be changed by accident while the program runs.

Alternative: `const double VAT_RATE = 0.07;`, a typed, read-only variable.

Q7 · Shop bill program

```c
#include <stdio.h>
#define VAT_RATE       0.07
#define DISCOUNT_RATE  0.05
#define DISCOUNT_LIMIT 2000.0
int main(void)
{
    double bill, discount = 0.0, vat, total;
    printf("Enter bill amount (Baht): ");
    scanf("%lf", &bill);
    if (bill > DISCOUNT_LIMIT)
        discount = bill * DISCOUNT_RATE;
    vat   = (bill - discount) * VAT_RATE;
    total = bill - discount + vat;
    printf("Bill     : %10.2f\n", bill);
    printf("Discount : %10.2f\n", discount);
    printf("VAT 7%%   : %10.2f\n", vat);
    printf("Total    : %10.2f\n", total);
    return 0;
}
```

Assumption: discount first, then VAT on the discounted price.

```
Enter bill amount (Baht): 2500
Bill     :    2500.00
Discount :     125.00
VAT 7%   :     166.25
Total    :    2541.25
```

```
Enter bill amount (Baht): 2000
Bill     :    2000.00
Discount :       0.00
VAT 7%   :     140.00
Total    :    2140.00
```

2000 gets no discount: “above” means >, not >=. Note %% prints a single %.

Q11 · Compound interest

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```c
#include <stdio.h>
#include <math.h>          /* for pow() */

int main(void)
{
    double P, R, T, A;

    printf("Principal (P): ");
    scanf("%lf", &P);
    printf("Rate %% per year (R): ");
    scanf("%lf", &R);
    printf("Time in years (T): ");
    scanf("%lf", &T);

    A = P * pow(1 + R / 100, T);

    printf("Final amount   = %.2f\n", A);
    printf("Interest earned = %.2f\n", A - P);
    return 0;
}
```

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    A = P * pow(1 + R / 100, T);

    printf("Final amount   = %.2f\n", A);
    printf("Interest earned = %.2f\n", A - P);
    return 0;
}
```

```
Principal (P): 10000
Rate % per year (R): 5
Time in years (T): 3
Final amount   = 11576.25
Interest earned = 1576.25
```

Check: $10000 \times 1.05^3 = 11576.25$ ✓

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Check: $10000 \times 1.05^3 = 11576.25$ ✓

Header: `<math.h>`

C has no power operator (^ is XOR!). `pow(x, y)` computes x^y and is declared in `<math.h>`. Link with `-lm` on Linux.

Q12 · Final velocity, with a safety check

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Q12 · Final velocity, with a safety check

Question. $v = \sqrt{u^2 + 2as}$. Read u, a, s and compute v. Print an error message when the value under the square root is negative.

```c
#include <stdio.h>
#include <math.h>          /* for sqrt() */

int main(void)
{
    double u, a, s, inside;

    printf("Enter u, a, s: ");
    scanf("%lf %lf %lf", &u, &a, &s);

    inside = u * u + 2 * a * s;
    if (inside < 0)
    {
        printf("Error: u^2 + 2as = %.2f is negative,"
               " no real velocity.\n", inside);
        return 1;
    }
    printf("v = %.2f\n", sqrt(inside));
    return 0;
}
```

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    return 0;
}
```

```
Enter u, a, s: 10 2 24
v = 14.00
```

$100 + 2 \cdot 2 \cdot 24 = 196,\ \sqrt{196} = 14$ ✓

```
Enter u, a, s: 3 -2 10
Error: u^2 + 2as = -31.00 is negative, no real
  velocity.
```

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$100 + 2 \cdot 2 \cdot 24 = 196$, $\sqrt{196} = 14$ ✓

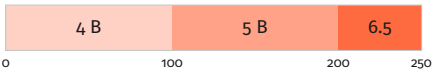
```
Enter u, a, s: 3 -2 10
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  velocity.
```

- Compute the inside **once**, test it, *then* call `sqrt`.
- `sqrt` of a negative number gives `nan` (not a number).
- `u * u` is simpler and exact; `pow(u, 2)` also works.

Q13 · Electricity bill: think in slabs

Question. First 100 units at 4 Baht, next 100 units at 5 Baht, every unit above 200 at 6.50 Baht. A fixed service charge of 25 Baht is added. Input a customer ID and units used; display the total bill.

Each slab is charged at its own rate:



Worked example: 250 units

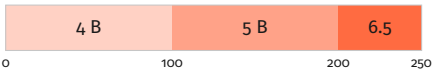
first 100 × 4	400.00

Mistake to avoid: charging *all* units at the highest rate (250×6.50). Only the units *above* 200 get 6.50.

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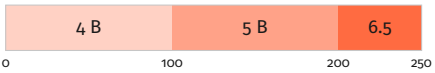
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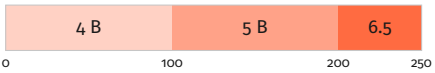
Worked example: 250 units

first 100 $\times$ 4	400.00
next 100 $\times$ 5	500.00
remaining 50 $\times$ 6.50	325.00

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Each slab is charged at its own rate:



Mistake to avoid: charging *all* units at the highest rate (250×6.50). Only the units *above* 200 get 6.50.

Worked example: 250 units

first 100 × 4	400.00
next 100 × 5	500.00
remaining 50 × 6.50	325.00
service charge	25.00
Total	**1250.00**

TIP Tiered electricity pricing: each band is charged at its own rate only, not the whole amount at the highest rate.

Q13 · Solution

```c
#include <stdio.h>

int main(void)
{
    int id;
    double units, bill;

    printf("Customer ID: ");
    scanf("%d", &id);
    printf("Units used : ");
    scanf("%lf", &units);

    if (units <= 100)
        bill = units * 4;
    else if (units <= 200)
        bill = 100 * 4 + (units - 100) * 5;
    else
        bill = 100 * 4 + 100 * 5 + (units - 200) * 6.50;

    bill = bill + 25;                       /* service charge */

    printf("Customer %d: total bill = %.2f Baht\n", id, bill);
    return 0;
}
```

```
Customer ID: 1002
Units used : 80
Customer 1002: total bill = 345.00 Baht
```

```
Customer ID: 1003
Units used : 150
Customer 1003: total bill = 675.00 Baht
```

```
Customer ID: 1001
Units used : 250
Customer 1001: total bill = 1250.00 Baht
```

Test **one value in each slab**. Here:

$80 \cdot 4 + 25 = 345,$

$400 + 50 \cdot 5 + 25 = 675.$

Q14 · Courier fees: `switch` + `if`

Question. Use `switch` for the zone (1, 2 or 3) and `if` statements for the weight to calculate the delivery fee. Print an error message for an invalid zone code.

Weight	Zone 1	Zone 2	Zone 3
up to 1 kg	40	60	350
above 1 to 5 kg	70	110	800
above 5 to 10 kg	120	180	1500
above 10 kg	200	300	2500

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- `switch (zone)` picks the **column**. `default:` catches every invalid zone.

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- Inside each `case`, an `if / else if` ladder picks the **row**.

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- `switch (zone)` picks the **column**. `default:` catches every invalid zone.
- Inside each `case`, an `if / else if` ladder picks the **row**.
- Boundaries: "up to 1 kg" ⇒ `w <= 1`; "above 1 to 5" ⇒ `w <= 5` (we already know `w > 1`).

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- `switch (zone)` picks the **column**. `default:` catches every invalid zone.
- Inside each `case`, an `if / else if` ladder picks the **row**.
- Boundaries: "up to 1 kg" ⇒ `w <= 1`; "above 1 to 5" ⇒ `w <= 5` (we already know `w > 1`).
- Don't forget `break;` after each case!

Q14 · Solution

```c
#include <stdio.h>

int main(void)
{
    int zone;
    double w, fee;

    printf("Zone (1-3): ");
    scanf("%d", &zone);
    printf("Weight (kg): ");
    scanf("%lf", &w);

    switch (zone)
    {
        case 1:
            if (w <= 1)       fee = 40;
            else if (w <= 5)  fee = 70;
            else if (w <= 10) fee = 120;
            else              fee = 200;
            break;
        case 2:
            if (w <= 1)       fee = 60;
            else if (w <= 5)  fee = 110;
            else if (w <= 10) fee = 180;
            else              fee = 300;
            break;
        case 3:
            if (w <= 1)       fee = 350;
            else if (w <= 5)  fee = 800;
            else if (w <= 10) fee = 1500;
            else              fee = 2500;
            break;
        default:
            printf("Error: invalid zone code %d\n", zone);
            return 1;
    }
    printf("Delivery fee = %.0f Baht\n", fee);
    return 0;
}
```

Q14 · Test runs (including the edges)

```
Zone (1-3): 2
Weight (kg): 3.5
Delivery fee = 110 Baht
```

```
Zone (1-3): 3
Weight (kg): 12
Delivery fee = 2500 Baht
```

```
Zone (1-3): 1
Weight (kg): 1
Delivery fee = 40 Baht
```

```
Zone (1-3): 4
Weight (kg): 2
Error: invalid zone code 4
```

- **Exactly 1 kg** is “up to 1 kg”, so the fee is 40, not 70. Always test the boundary values.
- Improvement: check `w <= 0` as an invalid weight too.

TIP Always test the boundary values, e.g. 1, 5, 10 kg. That is where < vs <= mistakes happen most.

Q15 · One function, three ways

Question. Read x and find y: $y = x^2 - 1$ if $x < 5$; $y = 50$ if $x = 5$; $y = 2x + 3$ if $x > 5$. Solve using (a) nested `if`, (b) an else-if ladder, (c) the conditional operator `?:`.

(a) nested if

```
if (x < 5)
    y = x * x - 1;
else
{
    if (x == 5)
        y = 50;
    else
        y = 2 * x + 3;
}
```

Q15 · One function, three ways

Question. Read x and find y: $y = x^2 - 1$ if $x < 5$; $y = 50$ if $x = 5$; $y = 2x + 3$ if $x > 5$. Solve using (a) nested `if`, (b) an else-if ladder, (c) the conditional operator `?:`.

(a) nested if

```
if (x < 5)
    y = x * x - 1;
else
{
    if (x == 5)
        y = 50;
    else
        y = 2 * x + 3;
}
```

(b) else-if ladder

```
if (x < 5)
    y = x * x - 1;
else if (x == 5)
    y = 50;
else
    y = 2 * x + 3;
```

Q15 · One function, three ways

Question. Read x and find y: $y = x^2 - 1$ if $x < 5$; $y = 50$ if $x = 5$; $y = 2x + 3$ if $x > 5$. Solve using (a) nested `if`, (b) an else-if ladder, (c) the conditional operator `? :`.

(a) nested if

```
if (x < 5)
    y = x * x - 1;
else
{
    if (x == 5)
        y = 50;
    else
        y = 2 * x + 3;
}
```

(b) else-if ladder

```
if (x < 5)
    y = x * x - 1;
else if (x == 5)
    y = 50;
else
    y = 2 * x + 3;
```

(c) conditional operator

```
y = (x < 5)  ? x * x - 1
  : (x == 5) ? 50
  :            2 * x + 3;
```

`c ? A : B` = A if c is true, else B.

Q15 · One function, three ways

Question. Read x and find y: $y = x^2 - 1$ if $x < 5$; $y = 50$ if $x = 5$; $y = 2x + 3$ if $x > 5$. Solve using (a) nested `if`, (b) an else-if ladder, (c) the conditional operator `? :`.

(a) nested if

```
if (x < 5)
    y = x * x - 1;
else
{
    if (x == 5)
        y = 50;
    else
        y = 2 * x + 3;
}
```

(b) else-if ladder

```
if (x < 5)
    y = x * x - 1;
else if (x == 5)
    y = 50;
else
    y = 2 * x + 3;
```

(c) conditional operator

```
y = (x < 5)  ? x * x - 1
  : (x == 5) ? 50
  :            2 * x + 3;
```

`c ? A : B` = A if c is true, else B.

```
x = 3
nested if: 8, ladder: 8, ?: 8
x = 5
nested if: 50, ladder: 50, ?: 50
x = 7
nested if: 17, ladder: 17, ?: 17
```

All three agree: $3^2 - 1 = 8$, 50, $2 \cdot 7 + 3 = 17$.
Note `x * x`, not `x^2`, and `==`, not `=`.

Q16 · The dangling `else`

Question. The indentation is misleading. (a) Which `if` owns the `else`? (b) Output for marks=40, att=90 and marks=60, att=70? (c) Rewrite with braces so "`Fail`" prints only when marks < 50.

```
if (marks >= 50)
    if (attendance >= 80)
        printf("Pass");
else
    printf("Fail");
```

Q16 · The dangling `else`

Question. The indentation is misleading. (a) Which `if` owns the `else`? (b) Output for marks=40, att=90 and marks=60, att=70? (c) Rewrite with braces so "`Fail`" prints only when marks < 50.

```
if (marks >= 50)
    if (attendance >= 80)
        printf("Pass");
else
    printf("Fail");
```

(a) An `else` pairs with the **nearest unmatched `if`**, here `if (attendance >= 80)`. Indentation means nothing to the compiler. The code really is:

```
if (marks >= 50) {
    if (attendance >= 80) printf("Pass");
    else                  printf("Fail");
}
```

Q16 · The dangling `else`

Question. The indentation is misleading. (a) Which `if` owns the `else`? (b) Output for marks=40, att=90 and marks=60, att=70? (c) Rewrite with braces so "`Fail`" prints only when marks < 50.

```
if (marks >= 50)
    if (attendance >= 80)
        printf("Pass");
else
    printf("Fail");
```

(a) An `else` pairs with the **nearest unmatched `if`**, here `if (attendance >= 80)`. Indentation means nothing to the compiler. The code really is:

```
if (marks >= 50) {
    if (attendance >= 80) printf("Pass");
    else                  printf("Fail");
}
```

(b)

marks=40, att=90	*(nothing)*	outer `if` is false, and it has no `else`
marks=60, att=70	`Fail`	outer true, inner false → its `else`

Q16 · (c) Fix it with braces

```c
if (marks >= 50)
{
    if (attendance >= 80)
        printf("Pass");
}
else
    printf("Fail");
```

The braces close the inner `if`, so the `else` can only belong to the outer `if (marks >= 50)`.

Real run of both versions (`q16_dangling.c`):

```
original(40, 90): []
original(60, 70): [Fail]
fixed(40, 90):    [Fail]
fixed(60, 70):    []
```

gcc even warns about the original:

```
warning: suggest explicit braces to avoid ambiguous 'else' [-Wdangling-else]
```

TIP An `else` pairs with the nearest unmatched `if`, regardless of indentation. Always use `{ }` to make it clear.

Q17 · A loop that multiplies

Question. Identify the initialization, test condition and update. Write the output, how many times the body runs, and the value of k after the loop.

```
for (k = 1; k <= 100; k = k * 3)
{
    printf("%d ", k);
}
```

Q17 · A loop that multiplies

Question. Identify the initialization, test condition and update. Write the output, how many times the body runs, and the value of k after the loop.

```
for (k = 1; k <= 100; k = k * 3)
{
    printf("%d ", k);
}
```

initialization	`k = 1` (runs once)
test condition	`k <= 100` (before every pass)
update	`k = k * 3` (after every pass)

Q17 · A loop that multiplies

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}
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initialization	`k = 1` (runs once)
test condition	`k <= 100` (before every pass)
update	`k = k * 3` (after every pass)

Trace

pass	k	k <= 100?	prints
1	1	yes	1

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{
    printf("%d ", k);
}
```

initialization	`k = 1` (runs once)
test condition	`k <= 100` (before every pass)
update	`k = k * 3` (after every pass)

Trace

pass	k	k <= 100?	prints
1	1	yes	1
2	3	yes	3

Q17 · A loop that multiplies

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```
for (k = 1; k <= 100; k = k * 3)
{
    printf("%d ", k);
}
```

initialization	`k = 1` (runs once)
test condition	`k <= 100` (before every pass)
update	`k = k * 3` (after every pass)

Trace

pass	k	k <= 100?	prints
1	1	yes	1
2	3	yes	3
3	9	yes	9

Q17 · A loop that multiplies

Question. Identify the initialization, test condition and update. Write the output, how many times the body runs, and the value of k after the loop.

```
for (k = 1; k <= 100; k = k * 3)
{
    printf("%d ", k);
}
```

initialization	`k = 1` (runs once)
test condition	`k <= 100` (before every pass)
update	`k = k * 3` (after every pass)

Trace

pass	k	k <= 100?	prints
1	1	yes	1
2	3	yes	3
3	9	yes	9
4	27	yes	27

Q17 · A loop that multiplies

Question. Identify the initialization, test condition and update. Write the output, how many times the body runs, and the value of k after the loop.

```
for (k = 1; k <= 100; k = k * 3)
{
    printf("%d ", k);
}
```

initialization	`k = 1` (runs once)
test condition	`k <= 100` (before every pass)
update	`k = k * 3` (after every pass)

Trace

pass	k	k <= 100?	prints
1	1	yes	1
2	3	yes	3
3	9	yes	9
4	27	yes	27
5	81	yes	81

Q17 · A loop that multiplies

Question. Identify the initialization, test condition and update. Write the output, how many times the body runs, and the value of k after the loop.

```
for (k = 1; k <= 100; k = k * 3)
{
    printf("%d ", k);
}
```

initialization	`k = 1` (runs once)
test condition	`k <= 100` (before every pass)
update	`k = k * 3` (after every pass)

Trace

pass	k	k <= 100?	prints
1	1	yes	1
2	3	yes	3
3	9	yes	9
4	27	yes	27
5	81	yes	81
—	**243**	no: stop	

Q17 · A loop that multiplies

Question. Identify the initialization, test condition and update. Write the output, how many times the body runs, and the value of k after the loop.

```
for (k = 1; k <= 100; k = k * 3)
{
    printf("%d ", k);
}
```

initialization	`k = 1` (runs once)
test condition	`k <= 100` (before every pass)
update	`k = k * 3` (after every pass)

Trace

pass	k	k <= 100?	prints
1	1	yes	1
2	3	yes	3
3	9	yes	9
4	27	yes	27
5	81	yes	81
—	**243**	no: stop	

```
1 3 9 27 81
body ran 5 times, k after loop = 243
```

Q18 · break stops everything

Question. Predict the output. Explain why the final value of `i` is not 11.

```
sum = 0;
for (i = 1; i <= 10; i++)
{
    sum = sum + i;
    if (sum > 20)
        break;
}
printf("i = %d, sum = %d", i, sum);
```

Q18 · break stops everything

Question. Predict the output. Explain why the final value of `i` is not 11.

```
sum = 0;
for (i = 1; i <= 10; i++)
{
    sum = sum + i;
    if (sum > 20)
        break;
}
printf("i = %d, sum = %d", i, sum);
```

i	sum after adding	sum > 20?
1	1	no

Q18 · break stops everything

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for (i = 1; i <= 10; i++)
{
    sum = sum + i;
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        break;
}
printf("i = %d, sum = %d", i, sum);
```

i	sum after adding	sum > 20?
1	1	no
2	3	no

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for (i = 1; i <= 10; i++)
{
    sum = sum + i;
    if (sum > 20)
        break;
}
printf("i = %d, sum = %d", i, sum);
```

i	sum after adding	sum > 20?
1	1	no
2	3	no
3	6	no

Q18 · break stops everything

Question. Predict the output. Explain why the final value of `i` is not 11.

```
sum = 0;
for (i = 1; i <= 10; i++)
{
    sum = sum + i;
    if (sum > 20)
        break;
}
printf("i = %d, sum = %d", i, sum);
```

i	sum after adding	sum > 20?
1	1	no
2	3	no
3	6	no
4	10	no

Q18 · break stops everything

Question. Predict the output. Explain why the final value of `i` is not 11.

```
sum = 0;
for (i = 1; i <= 10; i++)
{
    sum = sum + i;
    if (sum > 20)
        break;
}
printf("i = %d, sum = %d", i, sum);
```

i	sum after adding	sum > 20?
1	1	no
2	3	no
3	6	no
4	10	no
5	15	no

Q18 · break stops everything

Question. Predict the output. Explain why the final value of `i` is not 11.

```
sum = 0;
for (i = 1; i <= 10; i++)
{
    sum = sum + i;
    if (sum > 20)
        break;
}
printf("i = %d, sum = %d", i, sum);
```

i	sum after adding	sum > 20?
1	1	no
2	3	no
3	6	no
4	10	no
5	15	no
6	21	**yes** → break

Q18 · break stops everything

Question. Predict the output. Explain why the final value of `i` is not 11.

```
sum = 0;
for (i = 1; i <= 10; i++)
{
    sum = sum + i;
    if (sum > 20)
        break;
}
printf("i = %d, sum = %d", i, sum);
```

i	sum after adding	sum > 20?
1	1	no
2	3	no
3	6	no
4	10	no
5	15	no
6	21	**yes** → break

```
i = 6, sum = 21
```

Why not 11? `i` reaches 11 only if `i <= 10` becomes false. `break` exits **immediately**, skipping `i++`, so `i` stays 6.

Q19 · `continue` skips one pass

Question. Write the output. Explain how `continue` differs from `break` in this loop.

```
for (i = 1; i <= 15; i++)
{
    if (i % 2 == 0 || i % 5 == 0)
        continue;
    printf("%d ", i);
}
```

Q19 · `continue` skips one pass

Question. Write the output. Explain how `continue` differs from `break` in this loop.

```
for (i = 1; i <= 15; i++)
{
    if (i % 2 == 0 || i % 5 == 0)
        continue;
    printf("%d ", i);
}
```

Skip `i` if it is **even** or a **multiple of 5**:



Q19 · `continue` skips one pass

Question. Write the output. Explain how `continue` differs from `break` in this loop.

```
for (i = 1; i <= 15; i++)
{
    if (i % 2 == 0 || i % 5 == 0)
        continue;
    printf("%d ", i);
}
```

Skip `i` if it is **even** or a **multiple of 5**:




```
 1 3 7 9 11 13
```

Q19 · `continue` skips one pass

Question. Write the output. Explain how `continue` differs from `break` in this loop.

```
for (i = 1; i <= 15; i++)
{
    if (i % 2 == 0 || i % 5 == 0)
        continue;
    printf("%d ", i);
}
```

Skip `i` if it is **even** or a **multiple of 5**:




```
1 3 7 9 11 13
```

- `continue`: skip the **rest of this pass**, go to `i++`, and keep looping.
- `break`: leave the loop **completely**. With `break` here, the loop would stop at `i = 2` and print only `1`.

Q20 · Counting nested-loop executions (part 1)

Question. How many times will `printf()` execute?

```c
for (i = 0; i < 4; i++)
{
    for (j = i; j < 6; j++)
    {
        printf("#");
    }
}
```

i	j values	**times**

The inner loop runs for $j = i, i+1, \ldots, 5$, which is $6 - i$ values.

Q20 · Counting nested-loop executions (part 1)

Question. How many times will `printf()` execute?

```
for (i = 0; i < 4; i++)
{
    for (j = i; j < 6; j++)
    {
        printf("#");
    }
}
```

i	j values	**times**
0	0 1 2 3 4 5	6

The inner loop runs for $j = i, i+1, \ldots, 5$, which is $6 - i$ values.

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for (i = 0; i < 4; i++)
{
    for (j = i; j < 6; j++)
    {
        printf("#");
    }
}
```

i	j values	**times**
0	0 1 2 3 4 5	6
1	1 2 3 4 5	5

The inner loop runs for $j = i, i+1, \ldots, 5$, which is $6 - i$ values.

Q20 · Counting nested-loop executions (part 1)

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```
for (i = 0; i < 4; i++)
{
    for (j = i; j < 6; j++)
    {
        printf("#");
    }
}
```

The inner loop runs for $j = i, i+1, \ldots, 5$, which is $6 - i$ values.

i	j values	**times**
0	0 1 2 3 4 5	6
1	1 2 3 4 5	5
2	2 3 4 5	4

Q20 · Counting nested-loop executions (part 1)

Question. How many times will `printf()` execute?

```
for (i = 0; i < 4; i++)
{
    for (j = i; j < 6; j++)
    {
        printf("#");
    }
}
```

The inner loop runs for $j = i, i+1, \ldots, 5$, which is $6 - i$ values.

i	j values	**times**
0	0 1 2 3 4 5	6
1	1 2 3 4 5	5
2	2 3 4 5	4
3	3 4 5	3
	total	**18**

$6 + 5 + 4 + 3 = 18$. Verified by counting in a real run: `part 1: 18`.

Q20 · Counting nested-loop executions (part 2)

Question. Determine the total number of times `printf()` is executed. Show your working for each `i`.

```
for (i = 1; i <= 6; i++)
{
    for (j = 1; j <= i; j = j + 2)
    {
        printf("*");
    }
}
```

i	j values	**times**

`j` jumps by **2**: $1, 3, 5, \ldots$ while $j \leq i$. That's $\lceil i/2 \rceil$ values.

Q20 · Counting nested-loop executions (part 2)

Question. Determine the total number of times `printf()` is executed. Show your working for each `i`.

```
for (i = 1; i <= 6; i++)
{
    for (j = 1; j <= i; j = j + 2)
    {
        printf("*");
    }
}
```

i	j values	**times**
1	1	1

`j` jumps by **2**: $1, 3, 5, \ldots$ while $j \leq i$. That's $\lceil i/2 \rceil$ values.

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```
for (i = 1; i <= 6; i++)
{
    for (j = 1; j <= i; j = j + 2)
    {
        printf("*");
    }
}
```

i	j values	**times**
1	1	1
2	1	1

`j` jumps by **2**: $1, 3, 5, \ldots$ while $j \leq i$. That's $\lceil i/2 \rceil$ values.

Q20 · Counting nested-loop executions (part 2)

Question. Determine the total number of times `printf()` is executed. Show your working for each `i`.

```
for (i = 1; i <= 6; i++)
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    for (j = 1; j <= i; j = j + 2)
    {
        printf("*");
    }
}
```

`j` jumps by **2**: $1, 3, 5, \ldots$ while $j \leq i$. That's $\lceil i/2 \rceil$ values.

`i`	`j` values	**times**
1	1	1
2	1	1
3	1 3	2

Q20 · Counting nested-loop executions (part 2)

Question. Determine the total number of times `printf()` is executed. Show your working for each `i`.

```
for (i = 1; i <= 6; i++)
{
    for (j = 1; j <= i; j = j + 2)
    {
        printf("*");
    }
}
```

`j` jumps by **2**: $1, 3, 5, \ldots$ while $j \leq i$. That's $\lceil i/2 \rceil$ values.

`i`	`j` values	**times**
1	1	1
2	1	1
3	1 3	2
4	1 3	2

Q20 · Counting nested-loop executions (part 2)

Question. Determine the total number of times `printf()` is executed. Show your working for each `i`.

```
for (i = 1; i <= 6; i++)
{
    for (j = 1; j <= i; j = j + 2)
    {
        printf("*");
    }
}
```

`j` jumps by **2**: $1, 3, 5, \ldots$ while $j \leq i$. That's $\lceil i/2 \rceil$ values.

i	j values	**times**
1	1	1
2	1	1
3	1 3	2
4	1 3	2
5	1 3 5	3

Q20 · Counting nested-loop executions (part 2)

Question. Determine the total number of times `printf()` is executed. Show your working for each `i`.

```
for (i = 1; i <= 6; i++)
{
    for (j = 1; j <= i; j = j + 2)
    {
        printf("*");
    }
}
```

`j` jumps by **2**: $1, 3, 5, \ldots$ while $j \le i$. That's $\lceil i/2 \rceil$ values.

i	j values	**times**
1	1	1
2	1	1
3	1 3	2
4	1 3	2
5	1 3 5	3
6	1 3 5	3
	total	**12**

$1 + 1 + 2 + 2 + 3 + 3 = 12$ (real run: `part 2: 12`).

Q21 · Trace a nested loop

Question. Trace and write the complete output. How does the number of inner-loop executions change after each outer iteration?

```
for (i = 1; i <= 4; i++)
{
    for (j = 1; j <= 5 - i; j++)
    {
        printf("%d ", j);
    }
    printf("\n");
}
```

i	5 - i	inner runs	line printed

Q21 · Trace a nested loop

Question. Trace and write the complete output. How does the number of inner-loop executions change after each outer iteration?

```
for (i = 1; i <= 4; i++)
{
    for (j = 1; j <= 5 - i; j++)
    {
        printf("%d ", j);
    }
    printf("\n");
}
```

i	5 - i	inner runs	line printed
1	4	4	1 2 3 4

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for (i = 1; i <= 4; i++)
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    for (j = 1; j <= 5 - i; j++)
    {
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    }
    printf("\n");
}
```

i	5 - i	inner runs	line printed
1	4	4	1 2 3 4
2	3	3	1 2 3

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for (i = 1; i <= 4; i++)
{
    for (j = 1; j <= 5 - i; j++)
    {
        printf("%d ", j);
    }
    printf("\n");
}
```

i	5 - i	inner runs	line printed
1	4	4	1 2 3 4
2	3	3	1 2 3
3	2	2	1 2

Q21 · Trace a nested loop

Question. Trace and write the complete output. How does the number of inner-loop executions change after each outer iteration?

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for (i = 1; i <= 4; i++)
{
    for (j = 1; j <= 5 - i; j++)
    {
        printf("%d ", j);
    }
    printf("\n");
}
```

i	5 - i	inner runs	line printed
1	4	4	1 2 3 4
2	3	3	1 2 3
3	2	2	1 2
4	1	1	1

Q21 · Trace a nested loop

Question. Trace and write the complete output. How does the number of inner-loop executions change after each outer iteration?

```
for (i = 1; i <= 4; i++)
{
    for (j = 1; j <= 5 - i; j++)
    {
        printf("%d ", j);
    }
    printf("\n");
}
```

i	5 - i	inner runs	line printed
1	4	4	1 2 3 4
2	3	3	1 2 3
3	2	2	1 2
4	1	1	1

```
1 2 3 4
1 2 3
1 2
1
```

Inner loop: **one run fewer** per outer pass (4, 3, 2, 1), because its limit `5 - i` shrinks as `i` grows.

Q22 · `while` vs. `do-while`

Question. How many times is `*` printed in each, and what is the final `k`? What if `k` started at 3?

```c
int k = 0;
while (k > 0)
{
    printf("*");
    k--;
}
```

```c
int k = 0;
do
{
    printf("*");
    k--;
} while (k > 0);
```

Q22 · `while` vs. `do-while`

Question. How many times is `*` printed in each, and what is the final k? What if k started at 3?

```c
int k = 0;
while (k > 0)
{
    printf("*");
    k--;
}
```

```c
int k = 0;
do
{
    printf("*");
    k--;
} while (k > 0);
```

	start k = 0		start k = 3	
	`*` printed	final k	`*` printed	final k
`while`	0	0	3	0

Q22 · `while` vs. `do-while`

Question. How many times is `*` printed in each, and what is the final `k`? What if `k` started at 3?

```c
int k = 0;
while (k > 0)
{
    printf("*");
    k--;
}
```

```c
int k = 0;
do
{
    printf("*");
    k--;
} while (k > 0);
```

	start `k = 0`		start `k = 3`	
	`*` printed	final `k`	`*` printed	final `k`
`while`	0	0	3	0
`do-while`	**1**	−1	3	0

```
start k = 0
  while   : [] k = 0
  do-while: [*] k = -1
```

```
start k = 3
  while   : [***] k = 0
  do-while: [***] k = 0
```

`while` tests **before** the body, so it may run **0** times. `do-while` tests **after**, so it runs **at least once**. With `k = 3` they behave the same.

Q23 · Prefix and postfix `--`

Question. Find `a`, `b`, `c`. Then explain why an expression such as `a++ + a++` should never be used.

```
int a = 10;
int b = a--;
int c = --a;
```

`a--` (postfix)	**use** the old value, **then** decrease
`--a` (prefix)	**decrease** first, then use the new value

after line	a	b	c
`int a = 10;`	10	–	–

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after line	a	b	c
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`int b = a--;`	9	**10**	–

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```
a = 8, b = 10, c = 8
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`int b = a--;`	9	**10**	–
`int c = --a;`	**8**	10	**8**

```
a = 8, b = 10, c = 8
```

`b = a--`: b gets 10, *then* a becomes 9.

`c = --a`: a becomes 8 *first*, then c gets 8.

Q23 · Why a++ + a++ is forbidden

```c
int a = 5;
int x = a++ + a++;   /* ??? */
printf("x = %d, a = %d\n", x, a);
```

```
warning: operation on 'a' may be undefined
  [-Wsequence-point]
x = 11, a = 7
```

Our gcc printed `x = 11`. Is that “correct”? There **is no** correct answer.

TIP Never modify the same variable more than once in a single expression. The result is unpredictable (undefined behaviour).

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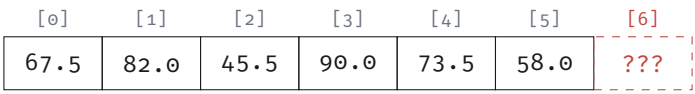
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- So it is **undefined behaviour**: other compilers or optimisation levels may give 10, 11, 12..., or anything else.
- **Rule**: modify a variable at most once per expression. Split it into separate statements.

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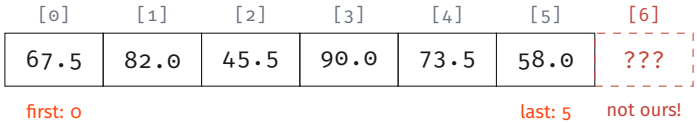
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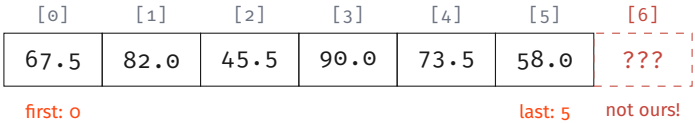
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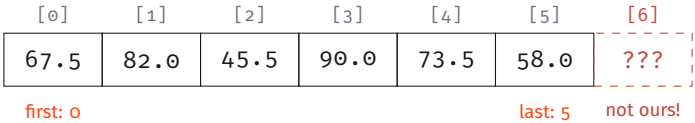
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1. Start with `high = low = marks[0]`, `sum = 0`.
2. One loop `i = 0 .. 5`: update `high` / `low`, add to `sum`.

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1. Start with `high = low = marks[0]`, `sum = 0`.
2. One loop `i = 0 .. 5`: update `high` / `low`, add to `sum`.
3. `avg = sum / 6`. A **second** loop counts `marks[i] > avg` (we need the average *before* we can compare).

Q24 · Solution

```c
#include <stdio.h>
#define N 6
int main(void)
{
    float marks[N] = {67.5, 82.0, 45.5, 90.0, 73.5, 58.0};
    float high = marks[0], low = marks[0], sum = 0, avg;
    int i, above = 0;
    for (i = 0; i < N; i++)
    {
        if (marks[i] > high) high = marks[i];
        if (marks[i] < low)  low  = marks[i];
        sum += marks[i];
    }
    avg = sum / N;
    for (i = 0; i < N; i++)
        if (marks[i] > avg)
            above++;
    printf("Highest = %.1f\n", high);
    printf("Lowest  = %.1f\n", low);
    printf("Average = %.2f\n", avg);
    printf("Above average: %d\n", above);
    return 0;
}
```

```
Highest = 90.0
Lowest  = 45.5
Average = 69.42
Above average: 3
```

sum $= 416.5$, avg $= 416.5/6 = 69.4167$

above it: 82.0, 90.0, 73.5 $\Rightarrow$ **3**

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#define N 6
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        sum += marks[i];
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    avg = sum / N;
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```

```
Highest = 90.0
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Average = 69.42
Above average: 3
```

sum $= 416.5$, avg $= 416.5/6 = 69.4167$

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`marks[6]`

Valid indices: **0 to 5**. `marks[6]` is past the end. C does **no bounds checking**: it compiles, but you read garbage, or a write corrupts other variables or crashes (undefined behaviour).

Q25 · The series $\sum 1/[n(n+1)]$

Question. $S = \sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ is evaluated with the loop below. (a) Why do the terms approach zero? (b) Show $S \to 1$. (c) Why `1.0` and not `1`? (d) How does the number of iterations grow with $1/\varepsilon$? Compare with terms like $1/2^n$.

```
n = 1;
sum = 0;
term = 1.0 / (n * (n + 1));
while (term > epsilon)
{
    sum = sum + term;
    n++;
    term = 1.0 / (n * (n + 1));
}
```

(a) Why the terms go to zero

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(a) Why the terms go to zero

n	$n(n+1)$	term
1	2	0.5
2	6	0.1667
3	12	0.0833
10	110	0.0091
100	10100	0.000099

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The denominator $n(n+1)$ grows without limit (roughly like n^2), while the numerator stays 1. So $\frac{1}{n(n+1)} \to 0$ as $n \to \infty$. The loop therefore stops eventually, for any $\varepsilon > 0$.

Q25 · (b) Why the sum approaches 1: telescoping

Split each term (partial fractions):

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1} \qquad \text{check: } \frac{(n+1)-n}{n(n+1)} = \frac{1}{n(n+1)} \checkmark$$

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Add the first N terms: almost everything cancels.

$$S_N = \left(1 - \tfrac{1}{2}\right) + \left(\tfrac{1}{2} - \tfrac{1}{3}\right) + \left(\tfrac{1}{3} - \tfrac{1}{4}\right) + \cdots + \left(\tfrac{1}{N} - \tfrac{1}{N+1}\right)$$

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Check with the program: after $N = 999$ terms, $S_{999} = 1 - \frac{1}{1000} = 0.999$, and the real run prints `sum = 0.999000`. ✓

TIP Telescoping = the middle terms all cancel out, leaving only the first and last.

Q25 · (c) Why `1.0` and not `1`?

`n * (n + 1)` is an `int`. With `1` on top, it's **integer division**: the fraction is thrown away.

`1 / (1 * 2)` $= 1/2 =$ **0**

`1.0 / (1 * 2)` $= 1.0/2 =$ **0.5**

```
1 / (n*(n+1)) with ints = 0,  1.0 / (n*(n+1)) = 0.50
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`1.0` is a `double`, so the `int` denominator is converted and the division keeps the fraction.

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With `1`: the very first term is `0`, so `0 > epsilon` is false. The loop **never runs** and the program reports `sum = 0`.

`1.0` is a `double`, so the `int` denominator is converted and the division keeps the fraction. Bonus: for very large n ($n > 46340$), `n * (n + 1)` itself overflows a 32-bit `int`. Writing `1.0 / ((double)n * (n + 1))` avoids that too.

Q25 · (d) How many iterations?

This series: the loop continues while

$$\frac{1}{n(n+1)} > \varepsilon \iff n(n+1) < \frac{1}{\varepsilon} \iff n \lesssim \sqrt{1/\varepsilon}$$

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Terms $1/2^n$**:**
$1/2^n > \varepsilon \iff 2^n < 1/\varepsilon \iff n < \log_2(1/\varepsilon)$,
so only **logarithmic** growth: 100× smaller ε costs just ≈ 7 more iterations.

Measured (`q25_series.c`):

ε	$1/[n(n+1)]$	$1/2^n$
10^{-2}	9	6
10^{-4}	99	13
10^{-6}	999	19

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10^{-6}	999	19

Matches the formulas: $\sqrt{10^6} = 1000$, $\log_2 10^6 \approx 19.9$.

Notice also: stopping at term $\leq \varepsilon$ leaves an error of $\frac{1}{N+1} \approx \sqrt{\varepsilon}$, *not* ε (0.999 for $\varepsilon = 10^{-6}$). Slowly converging series are both slow *and* less accurate than they look.

That's all 25

Top mistakes seen in this mock

- integer division (`1/2`, `total/n`)
- `=` vs `==`; `x^2` instead of `x*x`
- forgetting to initialize `sum`
- `<` vs `<=` at boundaries (Q7, Q13, Q14)
- dangling `else`, missing `break`
- array index 6 in an array of 6

Before the real exam

- Trace loops with a **table**, row by row.
- Test boundary inputs by hand.
- Write `%` specifiers carefully: `%d`, `%f`, `%lf`.
- Re-run every program in `examples/` and change the inputs.

TIP Good luck on the exam!